

Vijay Shankar M B

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SUMMARY

AI & Backend Engineer focused on building production-ready GenAI applications using Python, FastAPI, LangChain, LangGraph, RAG pipelines, vector databases, and scalable backend systems. Experienced in developing AI chat systems, document Q&A platforms, streaming architectures, semantic search pipelines, and full-stack AI products with modern deployment workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI / GenAI: LangChain, LangGraph, RAG, Vector Embeddings, Semantic Search, Prompt Engineering, LLM APIs, AI Agents, Retrieval Pipelines

Backend: FastAPI, Node.js, Express.js, REST APIs, Streaming APIs, JWT Authentication

Databases: PostgreSQL, MongoDB, ChromaDB, SQLAlchemy, Prisma ORM

Frontend: React, Next.js, Tailwind CSS

Developer Tools: Git, Postman, VS Code, Docker, Vercel, Render, GitHub

PROJECTS

AI Research Assistant

Python, FastAPI, LangChain, ChromaDB, PostgreSQL, Gemini API, Next.js, Tailwind CSS

- Built a production-style full-stack AI research assistant with real-time streaming AI chat, persistent conversation memory, and multi-session chat management.
- Developed PDF document Q&A functionality using Retrieval-Augmented Generation (RAG) pipelines with embeddings-based semantic search and vector retrieval using ChromaDB.
- Implemented document ingestion pipeline supporting PDF uploads, chunking strategies, vector embedding generation, and contextual retrieval for accurate AI responses.
- Engineered conversational memory and session management system using PostgreSQL to maintain persistent multi-session user conversations and chat history.
- Integrated source citation support to provide grounded AI responses with contextual document references.
- Built authentication system with secure user session handling and protected document access.
- Designed scalable FastAPI backend architecture with streaming responses and REST APIs for real-time AI interactions.
- Deployed production-ready frontend and backend with cloud database integration and modern deployment workflows.

ResumeIQ

Next.js, React 19, Express.js, Gemini API, MongoDB, NextAuth, Tailwind CSS

- Built an AI-powered resume analysis platform that parses PDF resumes and generates ATS, readability, and structure scores using custom evaluation pipelines.
- Integrated Gemini LLM APIs to generate AI-driven suggestions for summaries, skills, projects, and experience optimization.

- Developed resume-to-job-description comparison system with match scoring, missing skill detection, and actionable improvement recommendations.
- Implemented secure Google OAuth authentication with protected user sessions and cloud-based resume storage.
- Deployed frontend on Vercel and backend on Render with secure REST APIs, file upload handling, and schema validation.

EDUCATION

Diploma in Computer Application

JSS College, Mysore